

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Direct dual–channel E–stop inputs, watchdog–gated SR1 supply, separate RESET/ARM, redundant actuator interruption.

01 External listed sources and DC boundaries

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11 Watchdog PCB external connectors

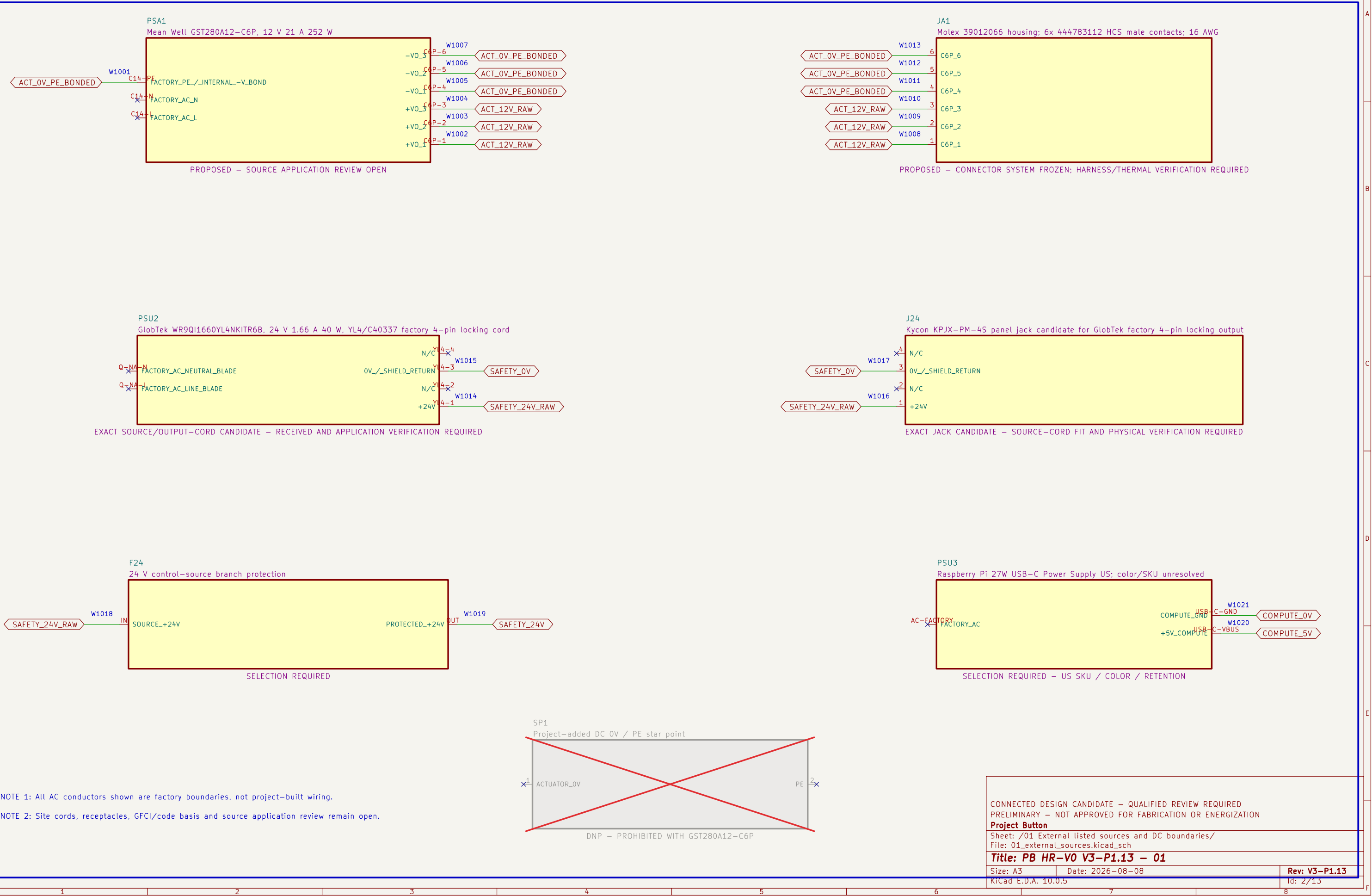
File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

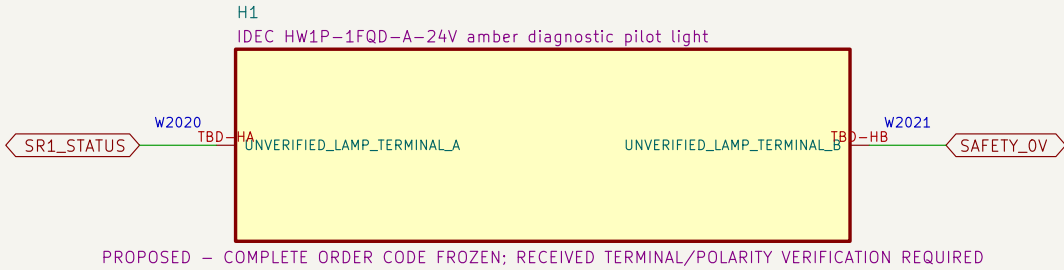
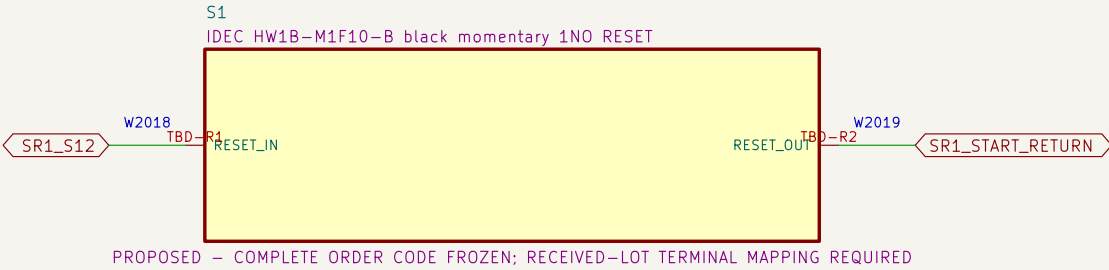
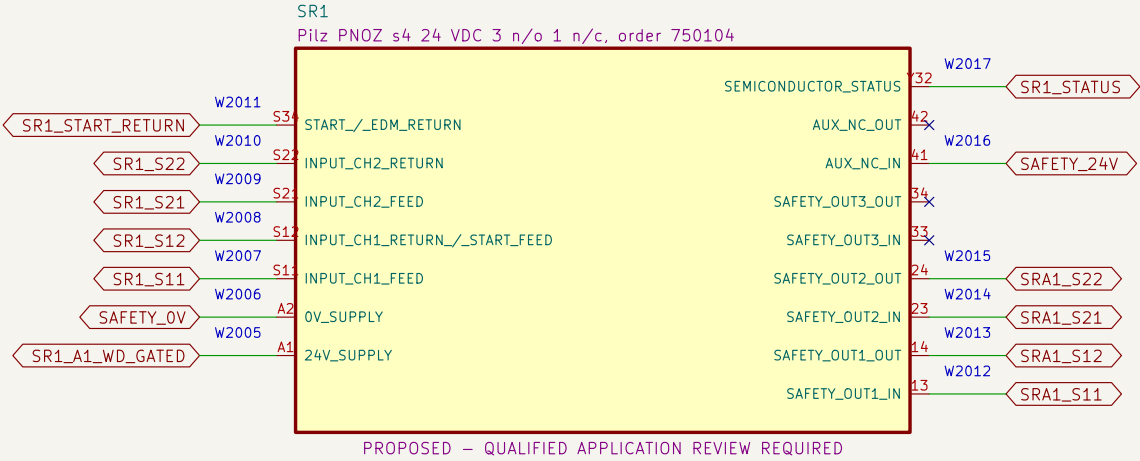
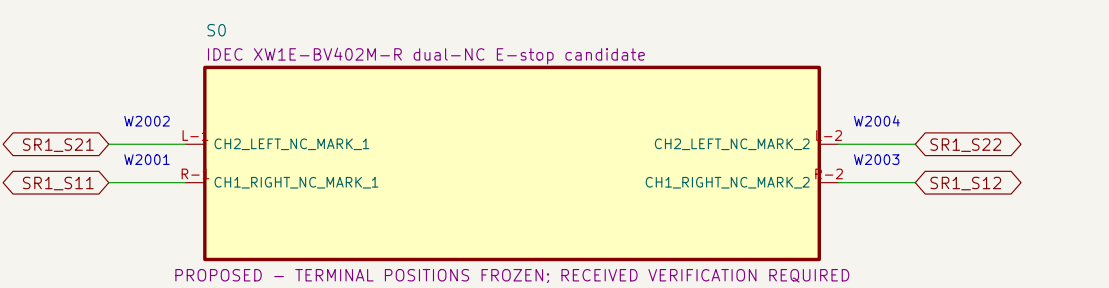
01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



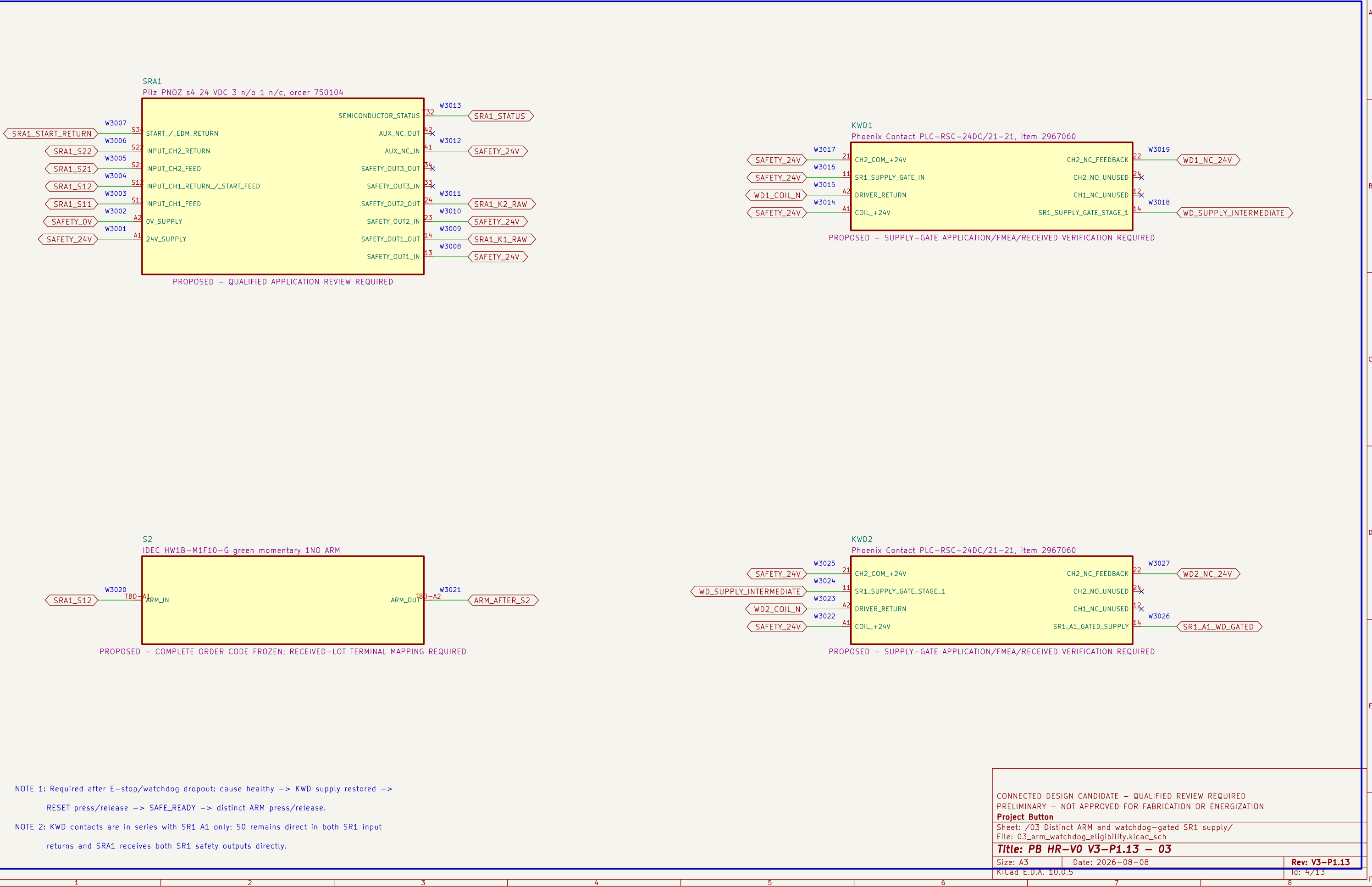
NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /02 Dual-channel E-stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and watchdog-gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.



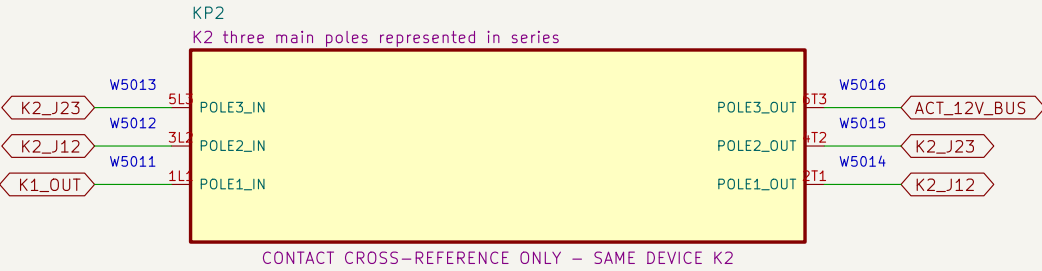
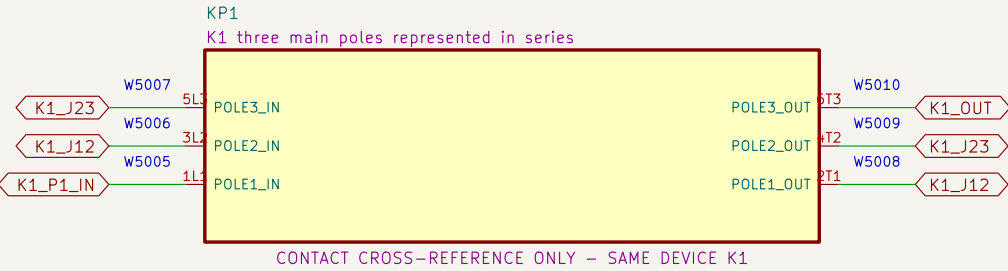
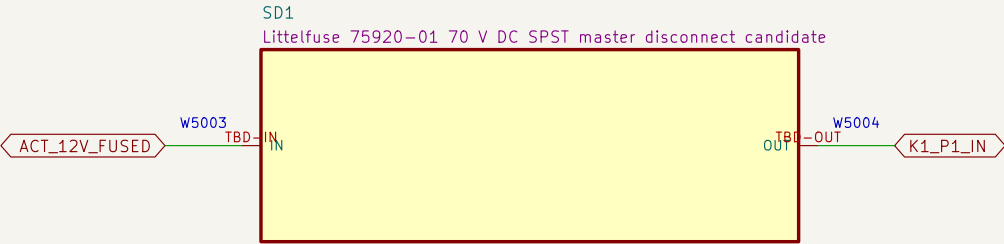
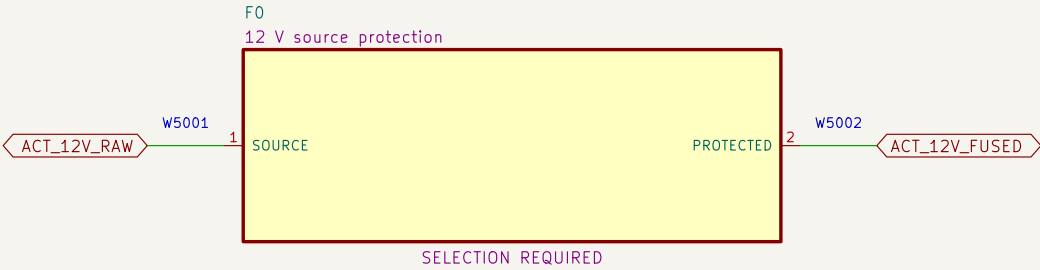
NOTE 1: Required after E-stop/watchdog dropout: cause healthy -> KWD supply restored ->
RESET press/release -> SAFE_READY -> distinct ARM press/release.

NOTE 2: KWD contacts are in series with SR1 A1 only; S0 remains direct in both SR1 input
returns and SRA1 receives both SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /03 Distinct ARM and watchdog-gated SR1 supply/
File: 03_arm_watchdog_eligibility.kicad_sch
Title: PB HR-V0 V3-P1.13 – 03
Size: A3
Date: 2026-08-08
Kicad E.D.A. 10.0.5
Rev: V3-P1.13
Id: 4/13

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



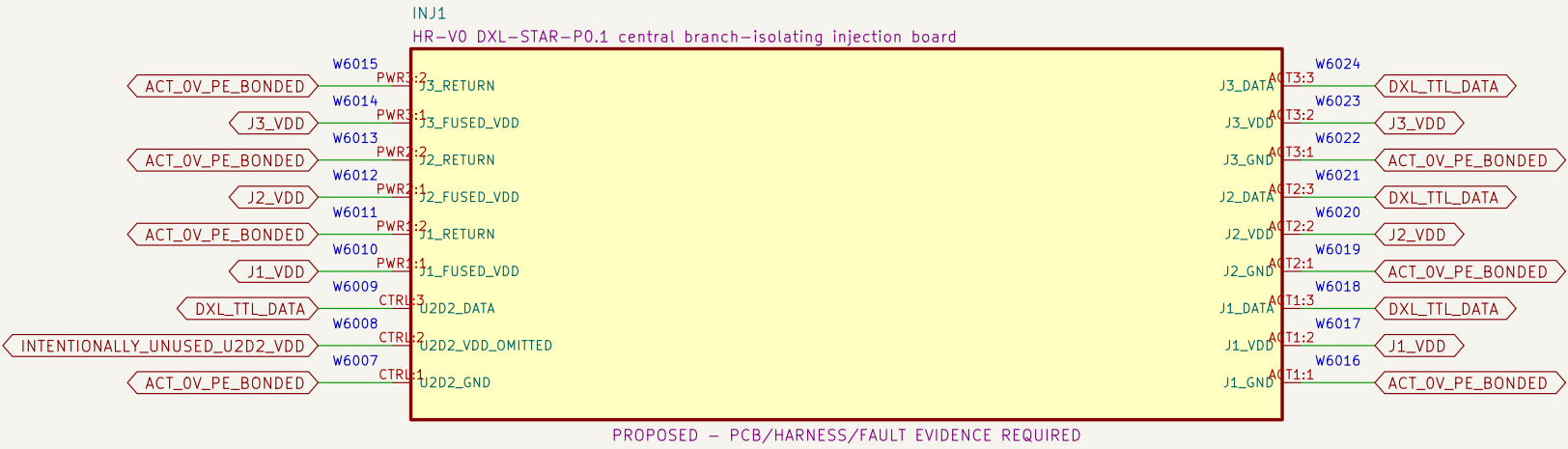
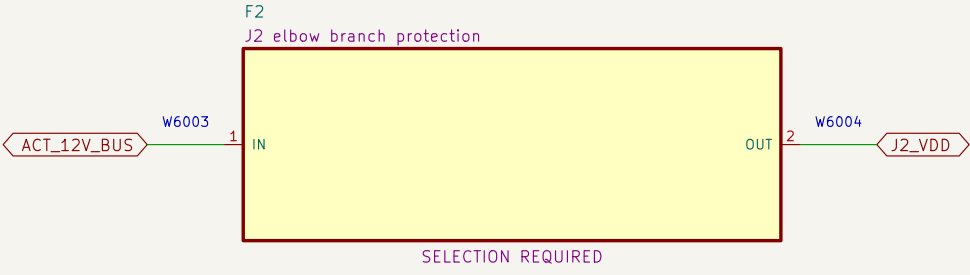
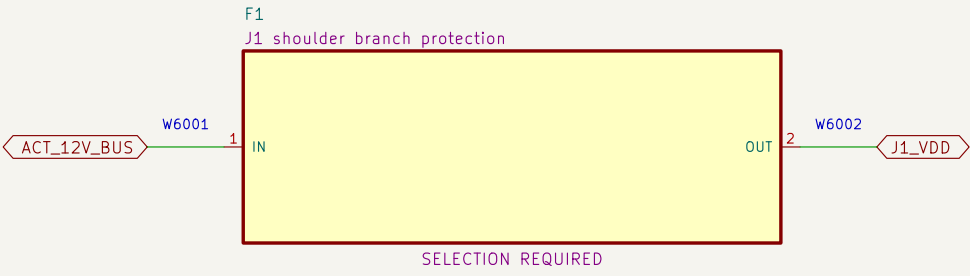
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–V0 V3–P1.13 – 05		
Size: A3	Date: 2026–08–08	Rev: V3–P1.13
KiCad E.D.A. 10.0.5		Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter-actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star; U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal-integrity, pull and no-backfeed tests remain release gates.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

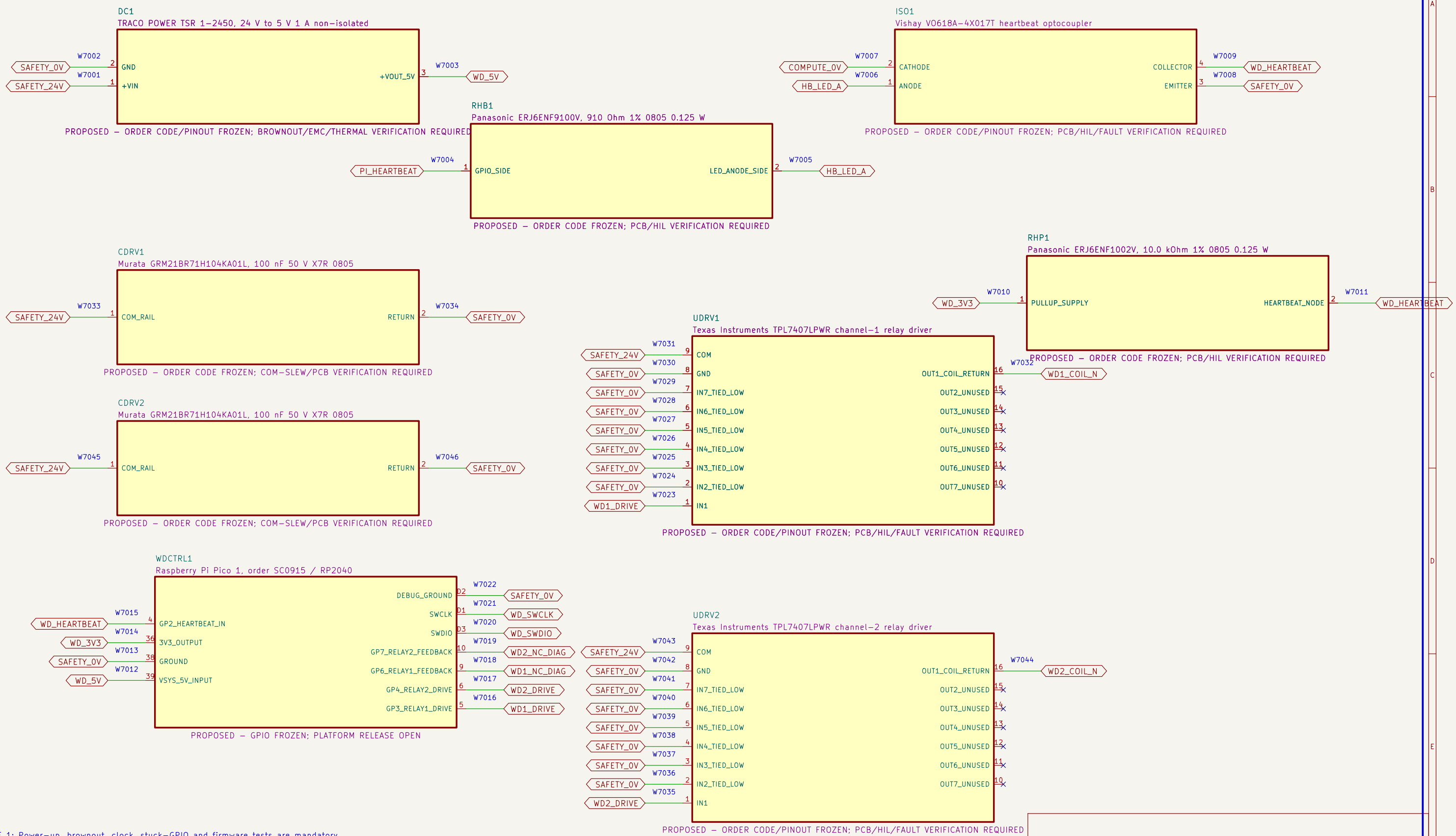
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Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
File: 06_branches_and_injection.kicad_sch

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Size: A3	Date: 2026-08-08	Rev: V3-P1.13
KiCad E.D.A. 10.0.5		Id: 7/13

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



NOTE 1: Power-up, brownout, clock, stuck-GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only;
current safety credit is NONE.

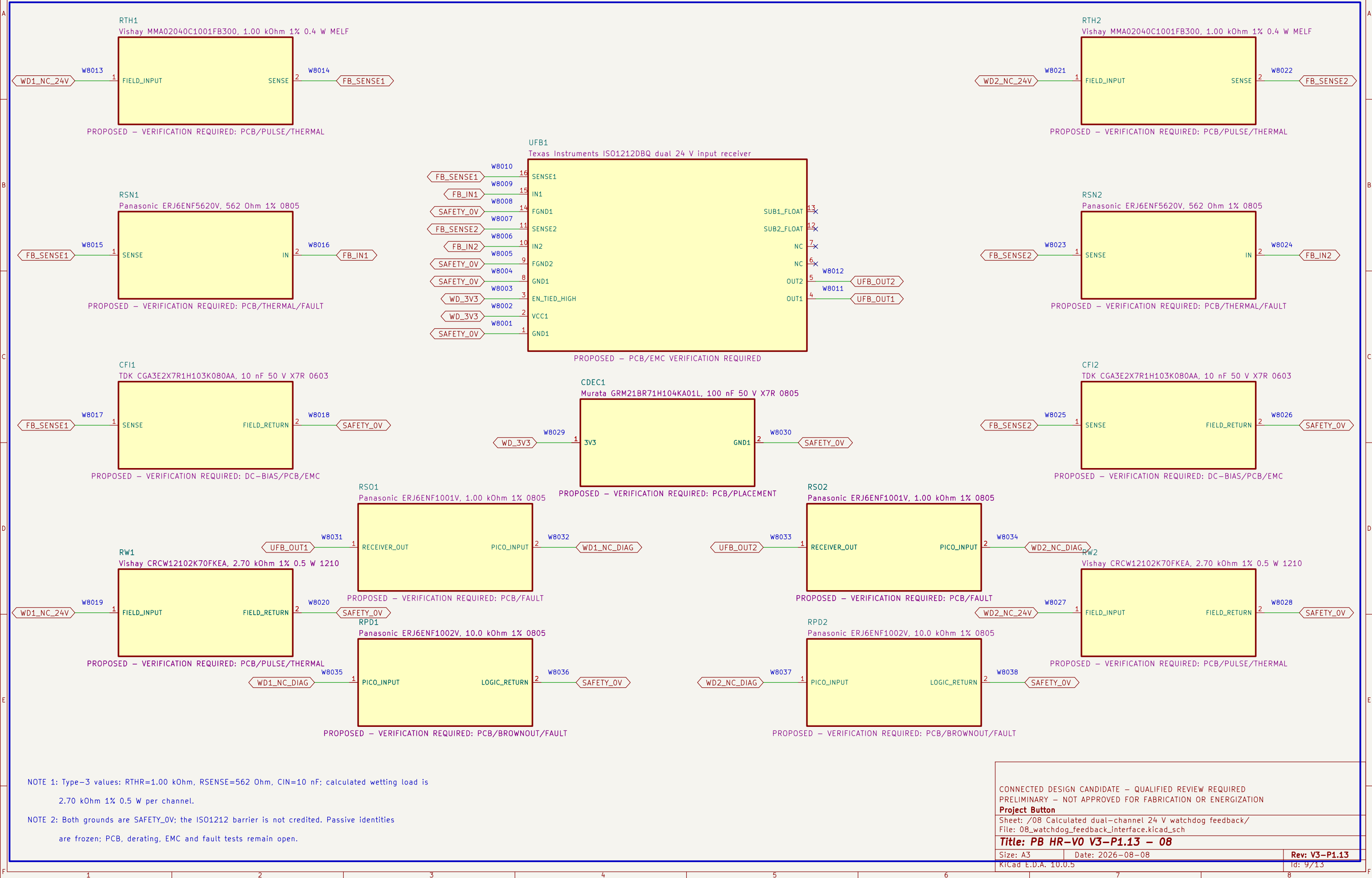
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad.sch

Title: PB HR-V0 V3-P1.13 - 07

Size: A3	Date: 2026-08-08	Rev: V3-P1.13
KICad E.D.A. 10.0.5		Id: 8/15

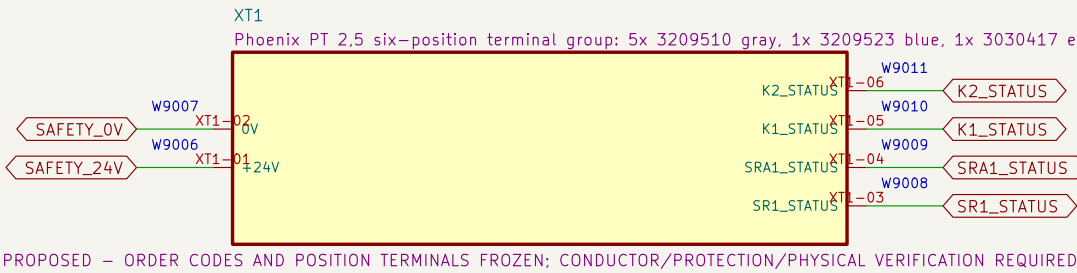
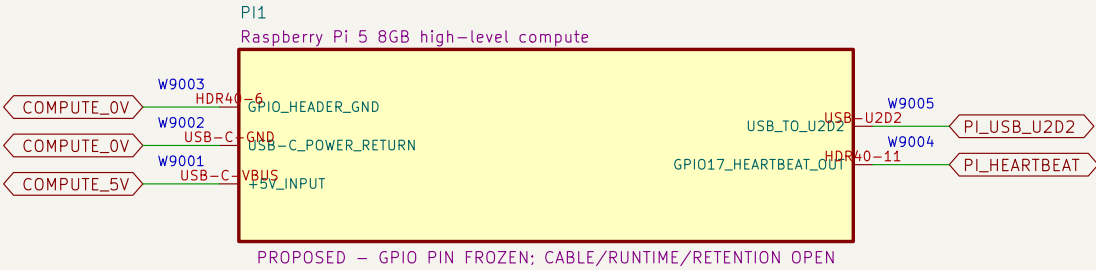
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button

Sheet: /09 Compute and control terminals/
File: 09_compute_and_control_terminals.kicad_sch

Title: PB HR-V0 V3-P1.13 – 09

Size: A3 Date: 2026-08-08

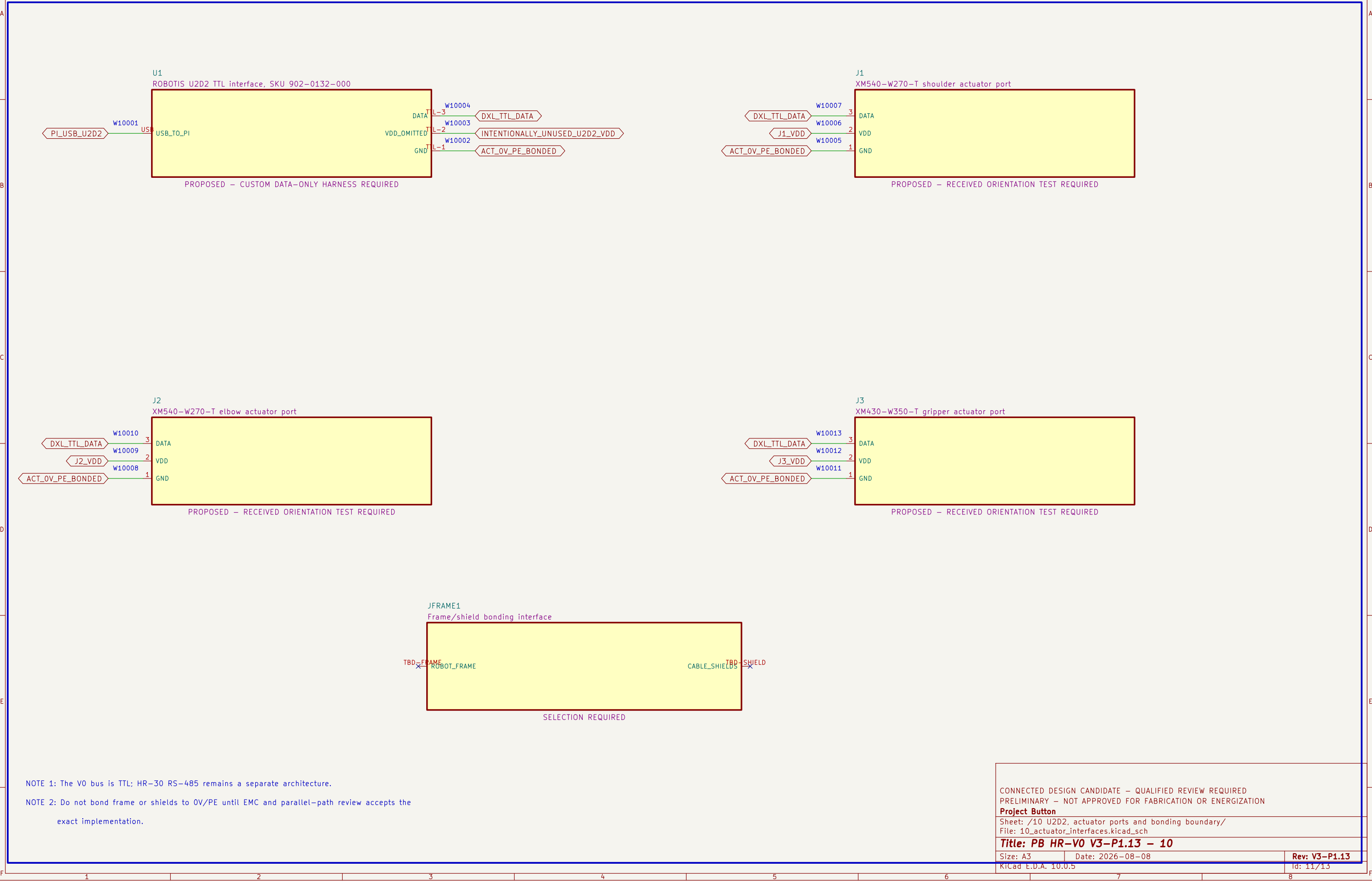
Rev: V3-P1.13

KiCad E.D.A. 10.0.5

Id: 10/13

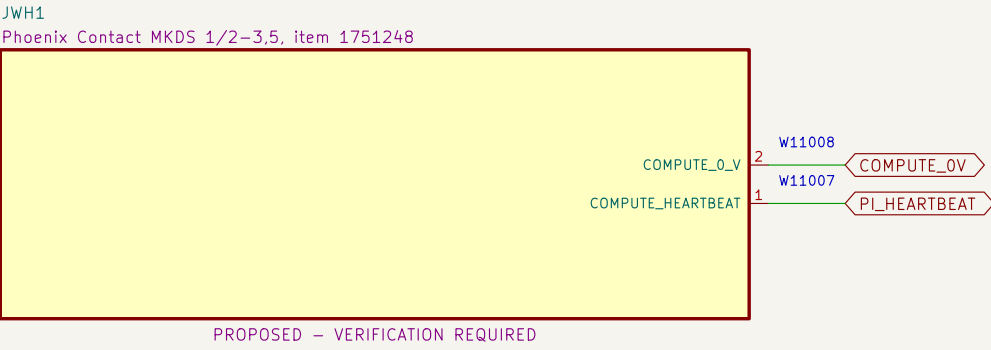
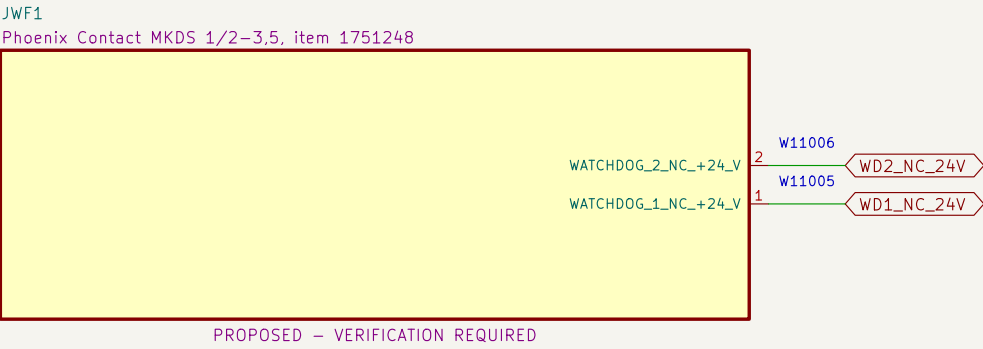
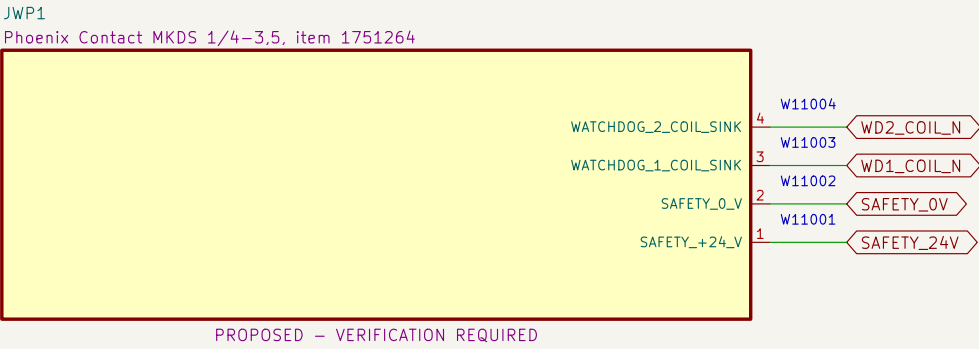
10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



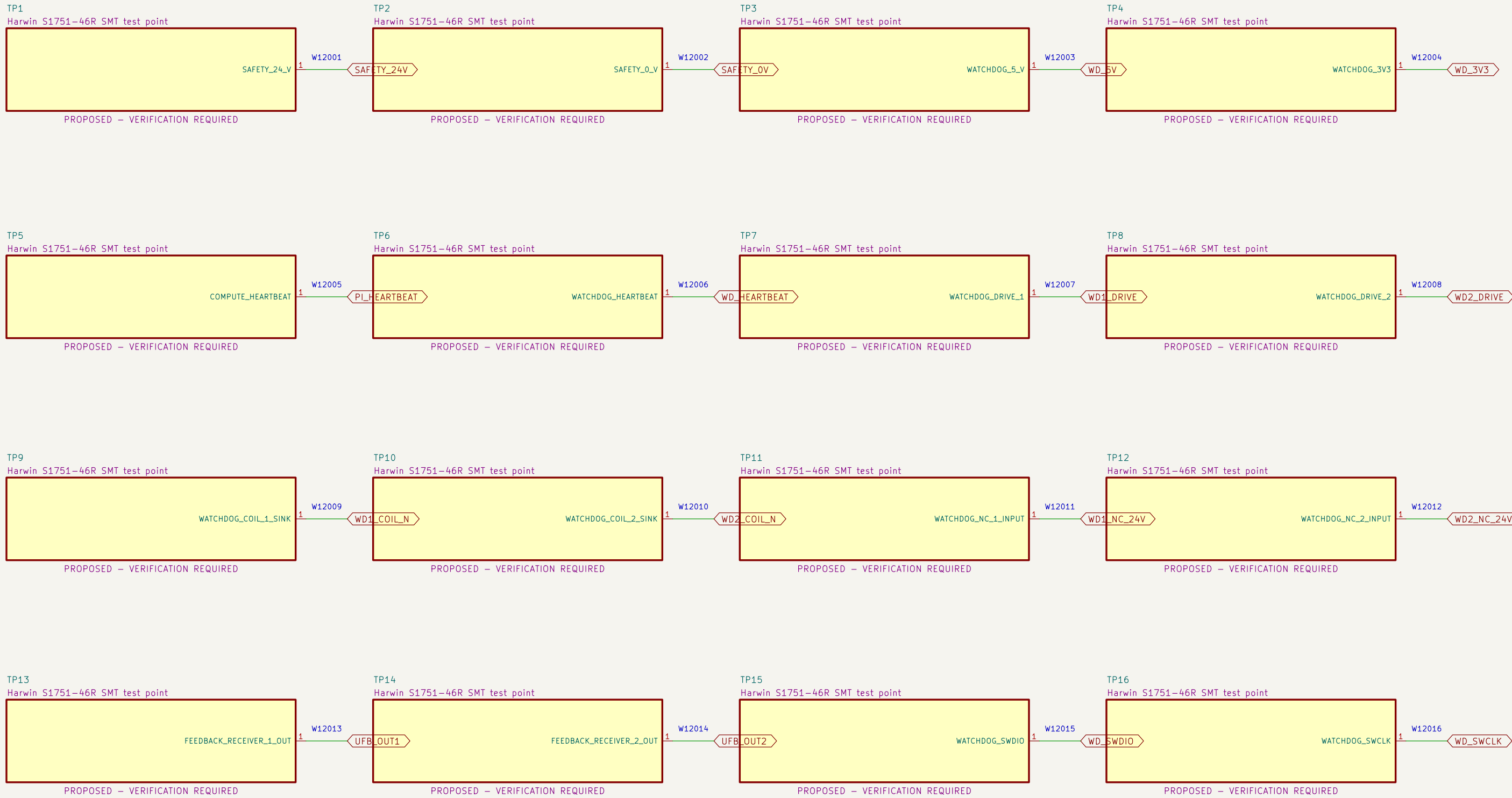
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR–V0 V3–P1.13 – 11		
Size: A3	Date: 2026–08–08	Rev: V3–P1.13
KiCad E.D.A. 10.0.5		Id: 12/13

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch

Title: PB HR-V0 V3-P1.13 – 12

Size: A3

Date: 2026-08-08

Rev: V3-P1.13

KiCad E.D.A. 10.0.5

Id: 13/13